AI-Based Optimization of Distillation Column Design

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Abstract

An AI-assisted optimization framework for distillation column design was developed using a combination of classical chemical engineering theory, rigorous process simulation, and machine learning.

The McCabe–Thiele method was first applied to estimate theoretical stage requirements for a benzene–toluene separation system. A rigorous column model was then constructed in DWSIM, from which a dataset was generated through sensitivity analysis by varying reflux ratio and number of stages.

Machine learning models were trained to approximate the nonlinear relationship between operating parameters and process outputs such as distillate purity and energy consumption. Bayesian optimization was applied to determine optimal operating conditions.

The predicted optimal design was finally validated through simulation, demonstrating the effectiveness of AI-based surrogate modeling for process design optimization.

1. Introduction

Distillation remains one of the most widely used separation processes in chemical engineering. Traditionally, distillation column design begins with simplified theoretical approaches such as the McCabe–Thiele graphical method, followed by rigorous simulation for accurate process evaluation.

Recent advances in artificial intelligence have enabled the integration of machine learning with classical process design methods. By training surrogate models on simulation data, complex design spaces can be explored more efficiently, enabling rapid optimization of operating parameters.

In distillation column design, a fundamental trade-off exists between **separation efficiency and energy consumption**. Achieving higher product purity generally requires either an increase in the reflux ratio or an increase in the number of theoretical stages in the column. Increasing the reflux ratio improves separation by returning a larger fraction of condensed liquid back to the column, thereby enhancing mass transfer between vapor and liquid phases. However, this simultaneously increases the **energy demand of both the condenser and the reboiler**, resulting in higher operational costs. Similarly, increasing the number of stages improves separation performance but leads to higher **capital costs and increased column complexity**.

Therefore, an optimal distillation column design must balance these competing objectives: **maximizing product purity while minimizing energy consumption and equipment requirements**. This trade-off makes distillation design a multi-objective optimization problem rather than a straightforward parameter selection task.

In this project, the classical McCabe–Thiele method is first used to obtain a theoretical baseline estimate of the column design. A rigorous distillation model is then constructed using DWSIM to generate simulation data under varying operating conditions. This dataset is subsequently used to train machine learning surrogate models that capture the relationship between key design variables—specifically **reflux ratio and number of stages**—and performance indicators such as **distillate purity and energy consumption**. Bayesian optimization is then applied to efficiently search the design space and identify operating conditions that achieve an improved balance between these competing objectives.

By integrating **theoretical analysis, rigorous simulation, and AI-based optimization**, the project demonstrates a modern approach to distillation design in which computational intelligence assists in identifying efficient operating strategies within complex process design spaces.

The objective of this project was to develop a hybrid workflow combining theoretical analysis, process simulation, and machine learning-based optimization to determine optimal distillation column operating conditions for the benzene–toluene separation system.

2. Project Objectives

The following objectives were established for the project:

- To perform theoretical distillation analysis using the McCabe–Thiele method.
- To construct and simulate a rigorous distillation column model in DWSIM.
- To generate a dataset through sensitivity analysis of operating parameters.
- To train machine learning surrogate models for predicting column performance.
- To apply Bayesian optimization to determine optimal reflux ratio and number of stages.
- To validate the predicted optimal design using rigorous simulation results.

3. Methodology Overview

The project methodology consisted of several sequential stages:

1. Theoretical analysis using McCabe–Thiele diagrams.
2. Rigorous distillation column simulation using DWSIM.
3. Dataset generation through sensitivity analysis.

4. Training of machine learning surrogate models.
5. Bayesian optimization of operating parameters.
6. Validation of the optimized design in the simulator.

This workflow enabled the integration of classical engineering principles with modern machine learning techniques.

4. McCabe–Thiele Theoretical Analysis

The McCabe–Thiele graphical method was used as an initial design estimate for the benzene–toluene separation.
Assuming constant relative volatility, the equilibrium curve and operating lines were constructed.
The feed condition was assumed to be a saturated liquid, resulting in a vertical q-line on the diagram.

The graphical stepping procedure between the equilibrium curve and operating lines allowed the approximate number of theoretical stages to be determined.
Although this method provides only an approximate estimate due to simplifying assumptions such as constant relative volatility and ideal equilibrium, it remains a valuable conceptual design tool.

5. Rigorous Distillation Simulation

A rigorous distillation column model was developed using DWSIM. The following steps were performed:

- Selection of benzene and toluene as components.
- Selection of an appropriate thermodynamic property package.
- Construction of the distillation column flowsheet.
- Definition of feed composition and operating conditions.
- Specification of reflux ratio and stage count.
- Execution of simulation and verification of convergence.

The rigorous simulation accounts for detailed thermodynamic behavior, mass balances, and energy balances, thereby providing a more realistic representation of the distillation process.

6. Dataset Generation Using Sensitivity Analysis

In order to train machine learning models, a dataset representing different operating conditions was required.
Sensitivity analysis in DWSIM was used to automatically generate simulation results for

multiple combinations of operating parameters.

Independent variables included:

- Reflux ratio
- Number of stages

Dependent variables recorded included:

- Distillate benzene purity
- Reboiler duty (energy consumption)

This dataset served as the training data for the machine learning models.

7. Machine Learning Surrogate Model

Machine learning regression models were trained to approximate the nonlinear relationship between operating variables and distillation performance.
Random forest regression was used due to its ability to capture nonlinear behavior and its robustness when working with relatively small datasets.

Two separate models were trained:

- A purity prediction model
- An energy consumption prediction model

These surrogate models were able to estimate column performance without requiring additional simulation runs.

8. Bayesian Optimization

Bayesian optimization was applied to identify optimal operating conditions.
Unlike brute-force grid search, Bayesian optimization iteratively explores the design space while balancing exploration and exploitation.

The optimization objective was defined as maximizing distillate purity while minimizing energy consumption.
The number of stages was constrained to integer values to ensure physically realistic column configurations.

The optimization algorithm proposed optimal combinations of reflux ratio and number of stages.

9. Validation of Optimized Design

The optimal design predicted by the machine learning model was validated using the DWSIM simulator.
The optimized reflux ratio and stage count were applied to the distillation column model, and the resulting purity and energy values were compared with the predictions of the surrogate model.

The results confirmed that the AI-assisted optimization approach was capable of identifying high-performance operating conditions.

10. Challenges Encountered During the Project

Several technical challenges were encountered during the development of the project.

Simulation convergence issues were initially encountered while working with ethanol–water systems due to azeotropic behavior.
This problem was resolved by switching to the benzene–toluene system, which exhibits more ideal vapor–liquid equilibrium behavior.

Difficulties were also encountered while configuring the distillation column specifications, feed stage location, and convergence parameters within DWSIM.
These issues were gradually resolved through adjustments in solver settings, specification definitions, and model configuration.

During the machine learning stage, dataset formatting issues were encountered while importing simulation data into Python.
These issues were resolved by cleaning the dataset structure and standardizing column headers.

Errors in McCabe–Thiele stage calculations were also observed due to incorrect implementation of the stage-stepping algorithm.
The algorithm was corrected to follow the proper horizontal–vertical stepping procedure.

11. Conclusions

The project successfully demonstrated how classical distillation design methods can be integrated with machine learning optimization techniques.
The McCabe–Thiele method provided an initial theoretical design estimate, while rigorous simulation produced accurate process data.
Machine learning surrogate models were able to approximate the behavior of the distillation column and significantly accelerate the optimization process.

The Bayesian optimization framework efficiently identified optimal operating conditions

that balance product purity and energy consumption.

This study illustrates the potential of AI-assisted methods for improving process design and optimization workflows in chemical engineering.